

“Power For AI Is A TRILLION-DOLLAR OPPORTUNITY”

An Indian startup is rearchitecting power delivery for AI servers, cutting energy losses, enhancing GPU performance, and developing chips poised to shape the next generation of global AI infrastructure. In a conversation with Preetam Tadeparthy and Vikram Gakhar of C2i Semiconductors, Nidhi Agarwal and Saba Aafreen from EFY examine the technology, challenges, and vision driving these innovations.



(L to R) C2i Semiconductors Founders—Vikram Gakhar, Founder & Director, Mixed Signal; Preetam Tadeparthy, Founder, CTO, VP, Engineering; Ram Anant, Founder & CEO; B.S. Dattatreya, Founder & Director, Power

What role does C2i Semiconductors play in powering next-generation AI infrastructure?

C2i Semiconductors stands for conversion, control, and intelligence, which define the company’s approach to power electronics. Conversion transforms electrical power between voltage domains, control regulates this process to ensure stability and efficiency, and intelligence enhances power management through embedded capabilities in

controllers or AI systems.

Building on this foundation, the company is developing a total power solution for AI infrastructure, particularly hyperscalers and data centres, covering the entire grid-to-core power delivery chain. **This includes converting the high-voltage grid input, typically 400 to 800 volts, into tightly regulated power suitable for advanced processors.** The company was founded with the ambition of building a product-driven

technology firm from India focused on rearchitecting end-to-end power delivery for server platforms while setting new benchmarks in innovation and quality for the global AI computing ecosystem.

What technical challenges has C2i Semiconductors faced in converting high-voltage input to sub-1-volt rails for AI processors?

Our first-generation product does not directly convert 800 volts to